

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the reduced selected-model TE Curve Verification Pipeline offline comparison for one dataset and one direction. It includes only the active selected model-development candidates plus reduced baseline anchors.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml` ;
- dataset root: `data\simplified_dataset` ;
- comparison mode: `reduced_selected_model_matrix` ;
- candidate count: 9 ;
- held-out curve count before candidate filtering: 97 ;
- percentage-error denominator: `peak_to_peak_truth` ;
- this reduced report excludes `global` candidates by design;
- only candidates valid for this direction are evaluated.

Candidate Inventory

Candidate	Family	Source	Kind	Surface	Valid Direction
polished_rcim_model_bank_reproduction_GBM19_Fw	GBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_rcim_model_bank_reproduction_ET19_Fw	ET	polished_rcim_model_bank_reproduction	track1_reference_bank	Fw	forward
polished_feedforward_Fw	feedforward	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_tree_Fw	tree	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_harmonic_regression_Fw	harmonic_regression	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_periodic_gru_sequence_Fw	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_periodic_mlp_harmonic_Fw	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	Fw	forward
polished_wave4_3_mixture_density_k3_Fw	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	Fw	forward
wave52b_offset_centered_shape_harmonic_Fw	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	Fw	forward

Forward Comparison

Wave 5.2B Offset And Harmonic Guided Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_Fw	0.058653	0.058748	128.839	229.455

Polished Model-Development Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_harmonic_regression_Fw	0.003230	0.003494	7.185	11.606
polished_tree_Fw	0.066315	0.066379	145.420	257.447
polished_wave4_3_mixture_density_k3_Fw	0.083190	0.083242	182.913	318.098
polished_periodic_mlp_harmonic_Fw	0.085830	0.085880	188.420	317.347
polished_periodic_gru_sequence_Fw	0.086270	0.086329	189.713	317.966
polished_feedforward_Fw	0.086931	0.087006	190.832	319.925

Polished RCIM Model-Bank Reproduction Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_rcim_model_bank_reproduction_GBM19_Fw	0.035161	0.035261	77.115	138.019
polished_rcim_model_bank_reproduction_ET19_Fw	0.038726	0.038810	85.150	149.396

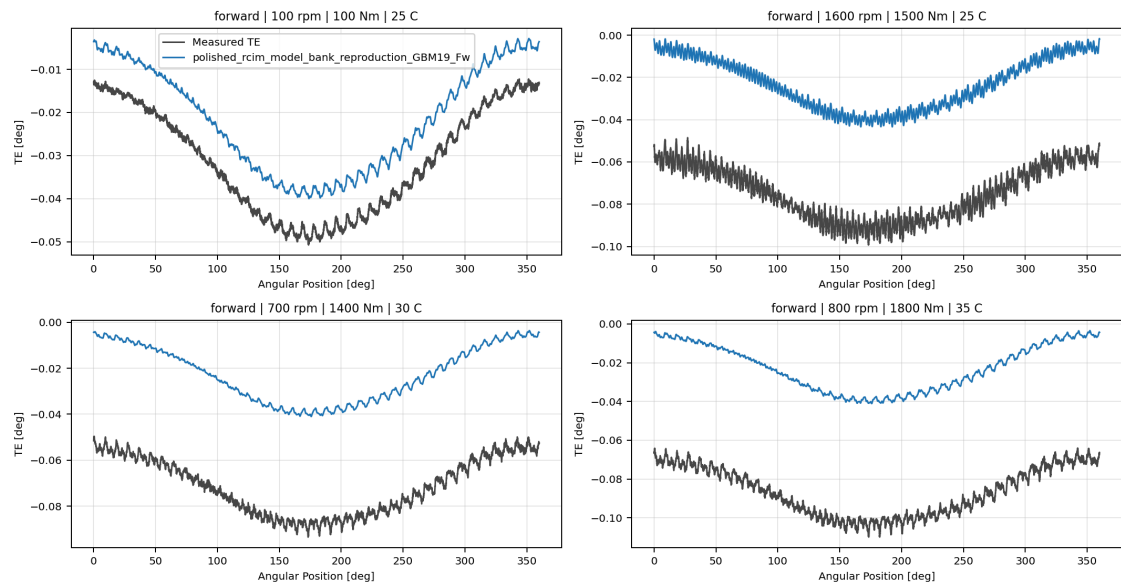
Curve Evidence

Each selected candidate is shown with four deterministic held-out curves. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_rcim_model_bank_reproduction_GBM19_Fw	Selected candidate	0.035161	0.035261	77.115	138.019
polished_rcim_model_bank_reproduction_ET19_Fw	Selected candidate	0.038726	0.038810	85.150	149.396
polished_feedforward_Fw	Selected candidate	0.086931	0.087006	190.832	319.925
polished_tree_Fw	Selected candidate	0.066315	0.066379	145.420	257.447
polished_harmonic_regression_Fw	Surface leader	0.003230	0.003494	7.185	11.606
polished_periodic_gru_sequence_Fw	Selected candidate	0.086270	0.086329	189.713	317.966
polished_periodic_mlp_harmonic_Fw	Selected candidate	0.085830	0.085880	188.420	317.347
polished_wave4_3_mixture_density_k3_Fw	Selected candidate	0.083190	0.083242	182.913	318.098
wave52b_offset_centered_shape_harmonic_Fw	Selected candidate	0.058653	0.058748	128.839	229.455

polished_rcim_model_bank_reproduction_GBM19_Fw

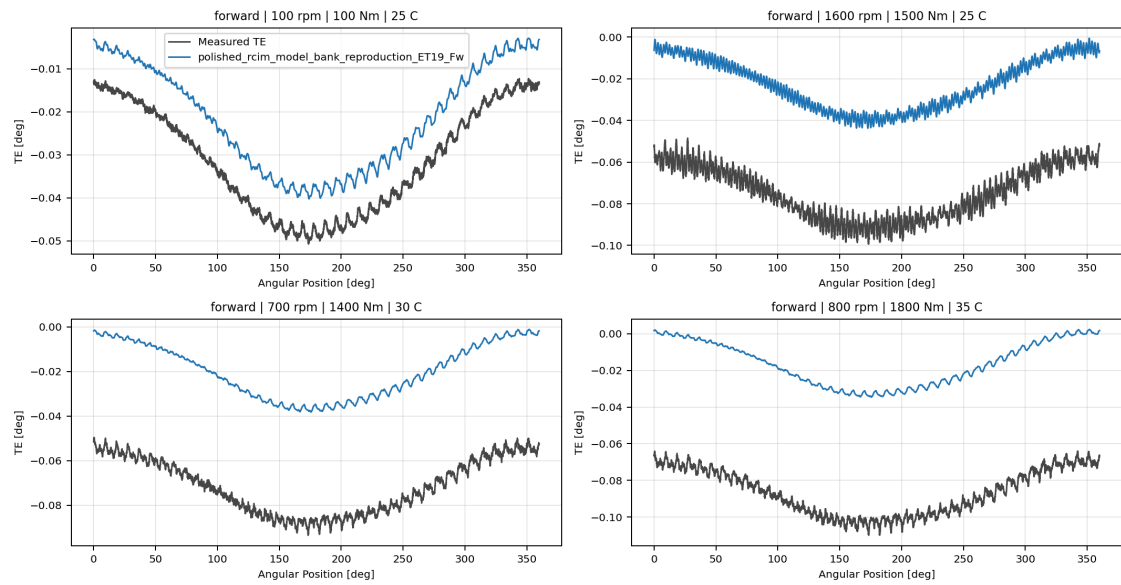
polished_rcim_model_bank_reproduction_GBM19_Fw



polished_rcim_model_bank_reproduction_GBM19_Fw measured-versus-predicted TE collage

polished_rcim_model_bank_reproduction_ET19_Fw

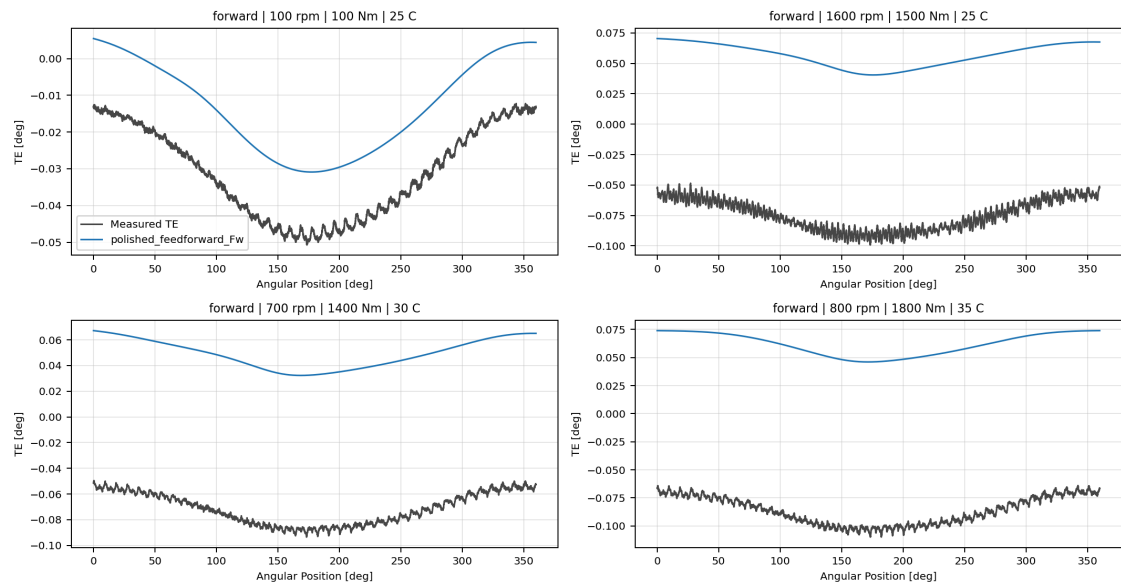
polished_rcim_model_bank_reproduction_ET19_Fw



polished_rcim_model_bank_reproduction_ET19_Fw measured-versus-predicted TE collage

polished_feedforward_Fw

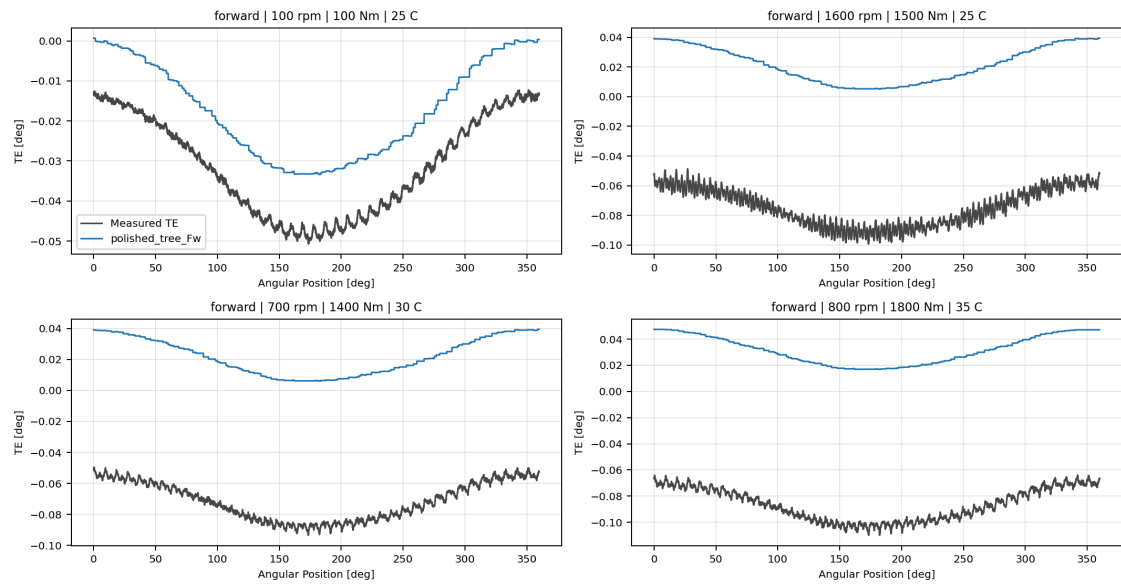
polished_feedforward_Fw



polished_feedforward_Fw measured-versus-predicted TE collage

polished_tree_Fw

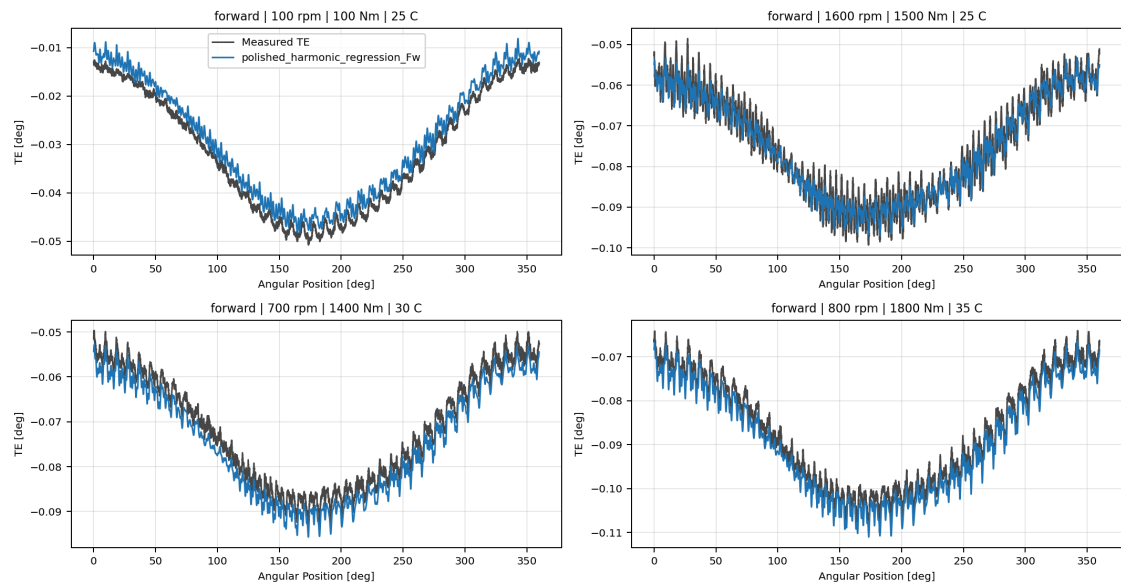
polished_tree_Fw



polished_tree_Fw measured-versus-predicted TE collage

polished_harmonic_regression_Fw

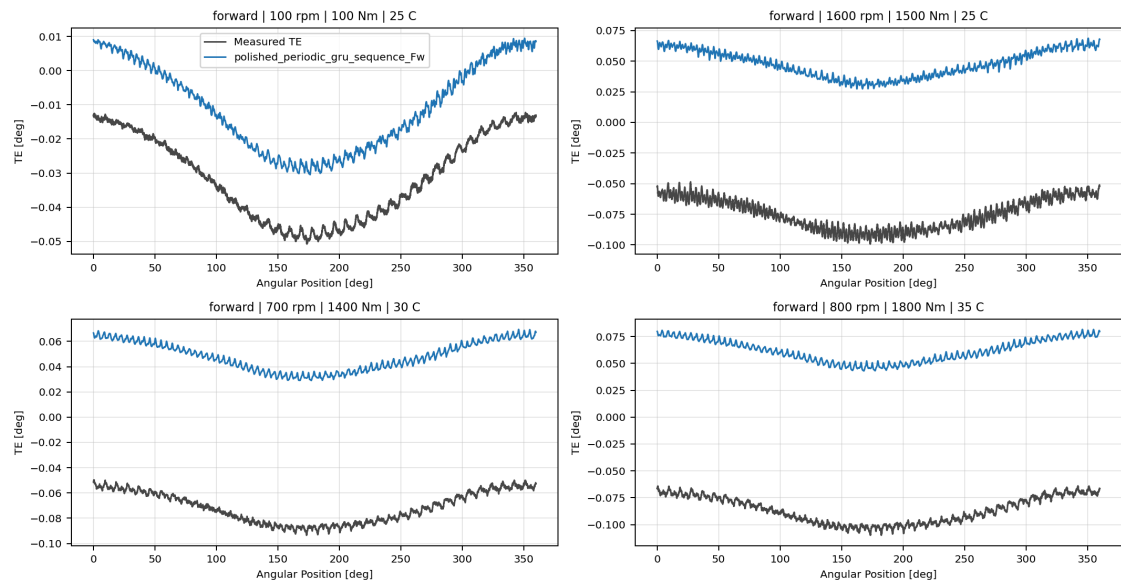
polished_harmonic_regression_Fw



polished_harmonic_regression_Fw measured-versus-predicted TE collage

polished_periodic_gru_sequence_Fw

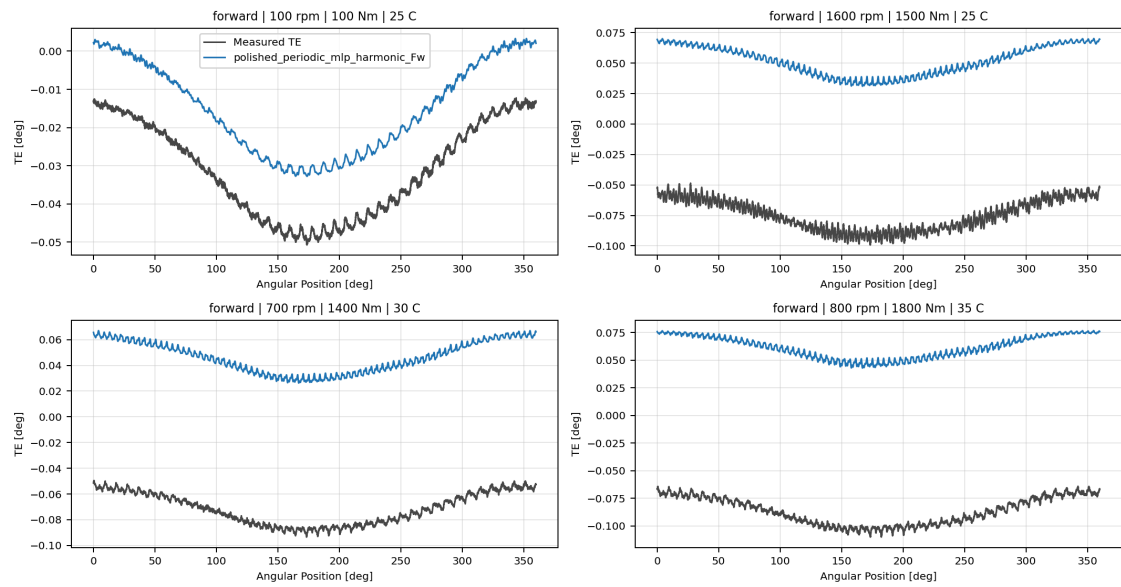
polished_periodic_gru_sequence_Fw



polished_periodic_gru_sequence_Fw measured-versus-predicted TE collage

polished_periodic_mlp_harmonic_Fw

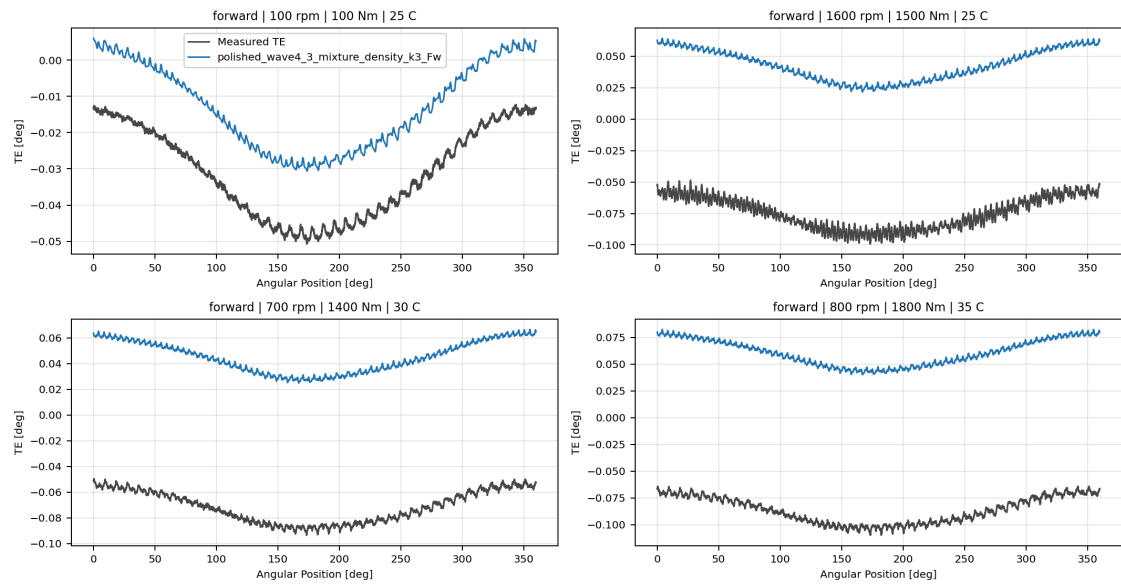
polished_periodic_mlp_harmonic_Fw



polished_periodic_mlp_harmonic_Fw measured-versus-predicted TE collage

polished_wave4_3_mixture_density_k3_Fw

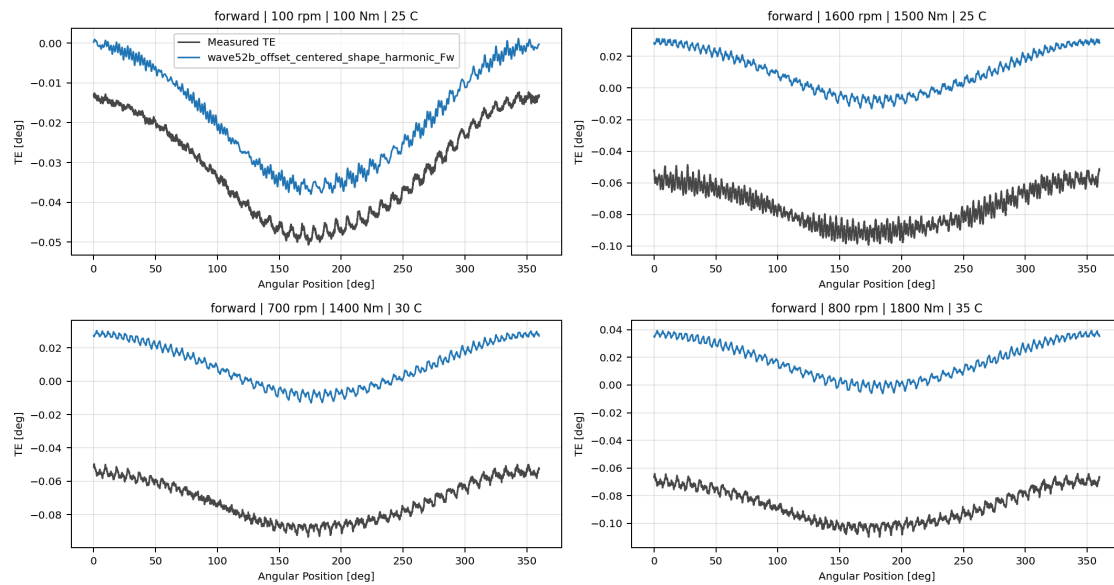
polished_wave4_3_mixture_density_k3_Fw



polished_wave4_3_mixture_density_k3_Fw measured-versus-predicted TE collage

wave52b_offset_centered_shape_harmonic_Fw

wave52b_offset_centered_shape_harmonic_Fw



wave52b_offset_centered_shape_harmonic_Fw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-06-19-53-27_track2_reduced_selected_model_matrix_track2_selected_simplified_dataset_forward_2026_07_06\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-06-19-53-27_track2_reduced_selected_model_matrix_track2_selected_simplified_dataset_forward_2026_07_06\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\paused_reduced_selected_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional candidates are never evaluated on the opposite direction in this reduced report. `global` is paused for the active reduced pipeline and is not included in the candidate inventory. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.